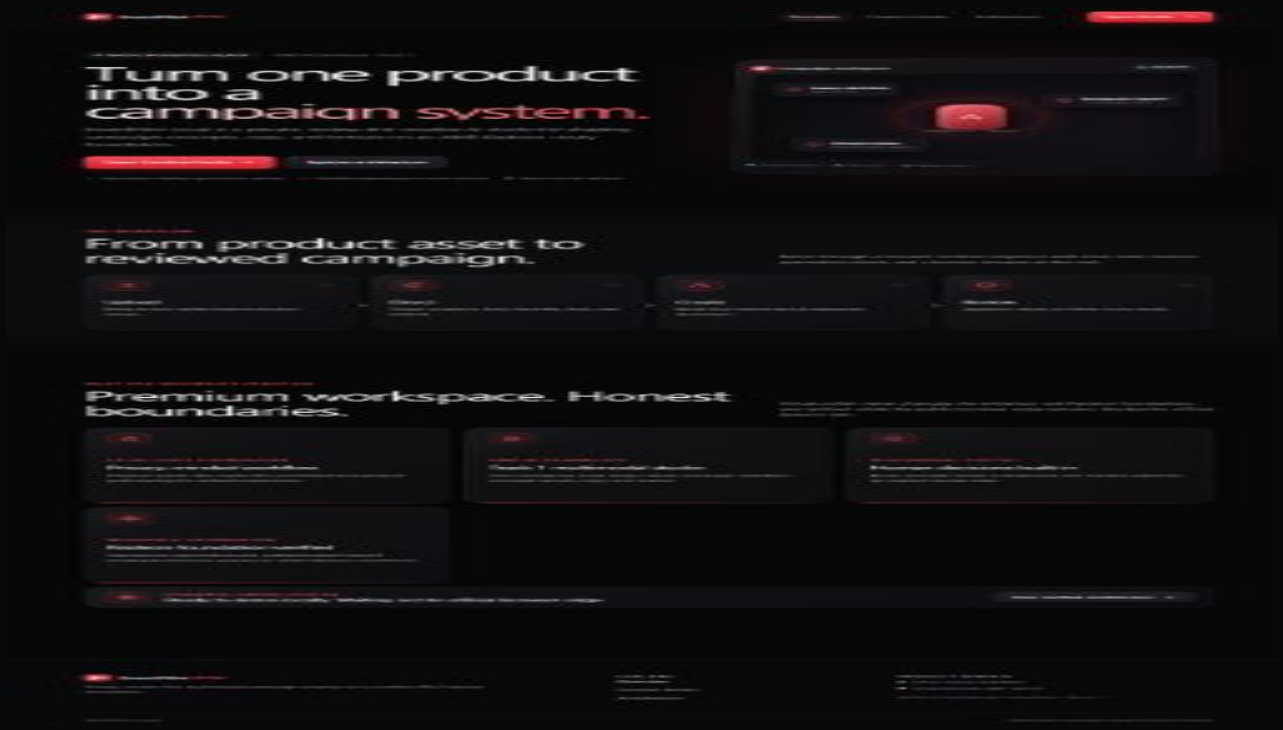
AMD AI DEVMASTER HACKATHON - TRACK 1

BrandPilot Local

A review-first campaign creation studio on a verified AMD Radeon and ROCm foundation.



PROJECT PROFILE

Talal Khawaja | Aqeela Urooj

Mock-first local product. Recorded Radeon evidence. Honest browser-edge boundary.

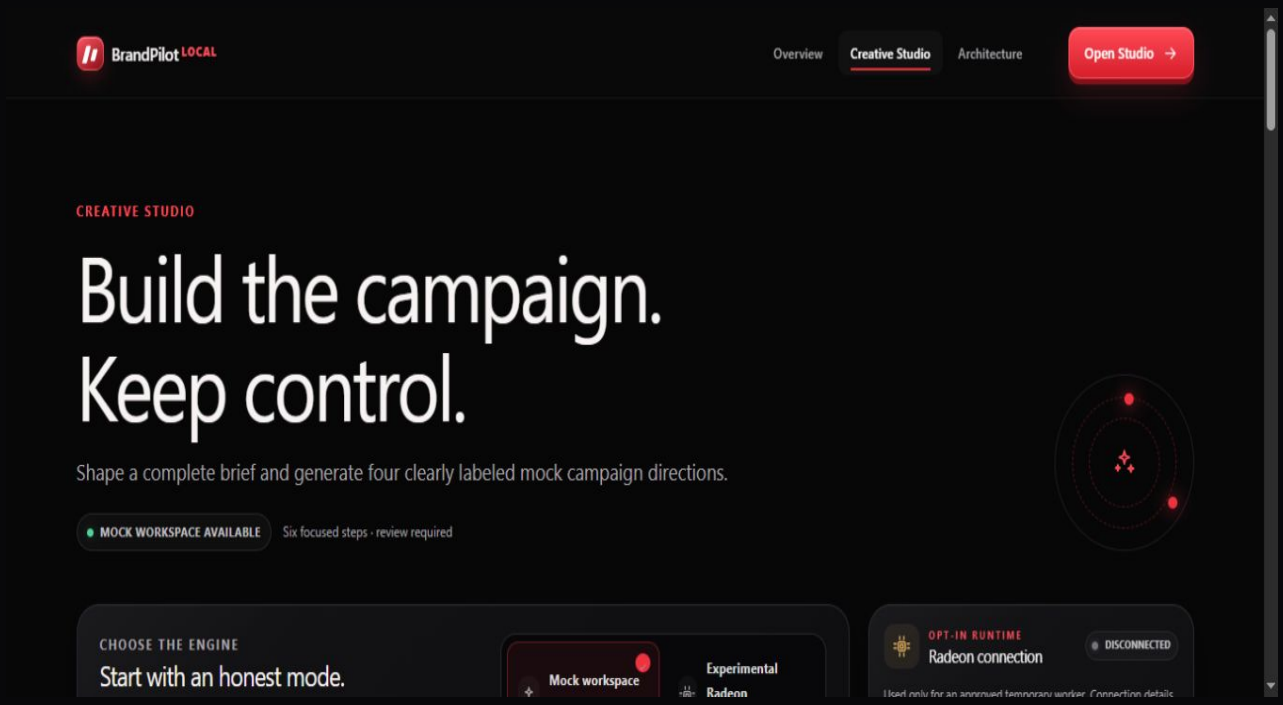
PROBLEM

The campaign workflow is fragmented

Small brands and creative teams often move one campaign across prompt tools, design files, copy documents, and approval threads. That fragmentation slows iteration, weakens traceability, and can expose sensitive product assets to unnecessary services.

BrandPilot brings the system together

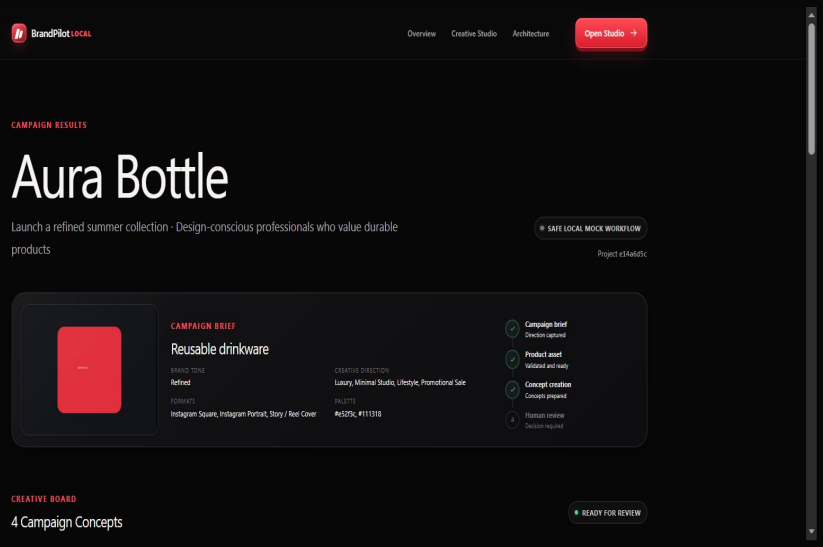
- Validated rights-cleared product-image intake
- Audience, benefits, tone, colors, creative direction, formats, and guardrails
- Multiple disclosed campaign concepts and copy in one board
- Persisted human approval, rejection, or needs-changes state
- Mock by default; authenticated Radeon worker only by explicit opt-in



PRODUCT EXPERIENCE

One asset becomes an accountable campaign

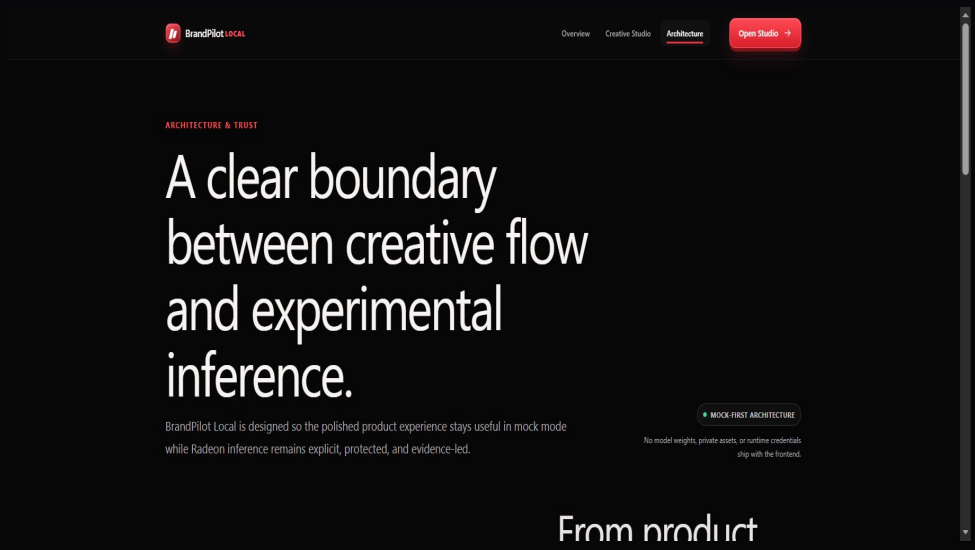
01	**Upload** Validated product image
02	**Direct** Campaign context and guardrails
03	**Create** Four labelled mock directions
04	**Review** Stored human decision



The screen above is a local mock demonstration. Mock concepts are explicitly labelled and do not imply GPU execution.

SYSTEM DESIGN

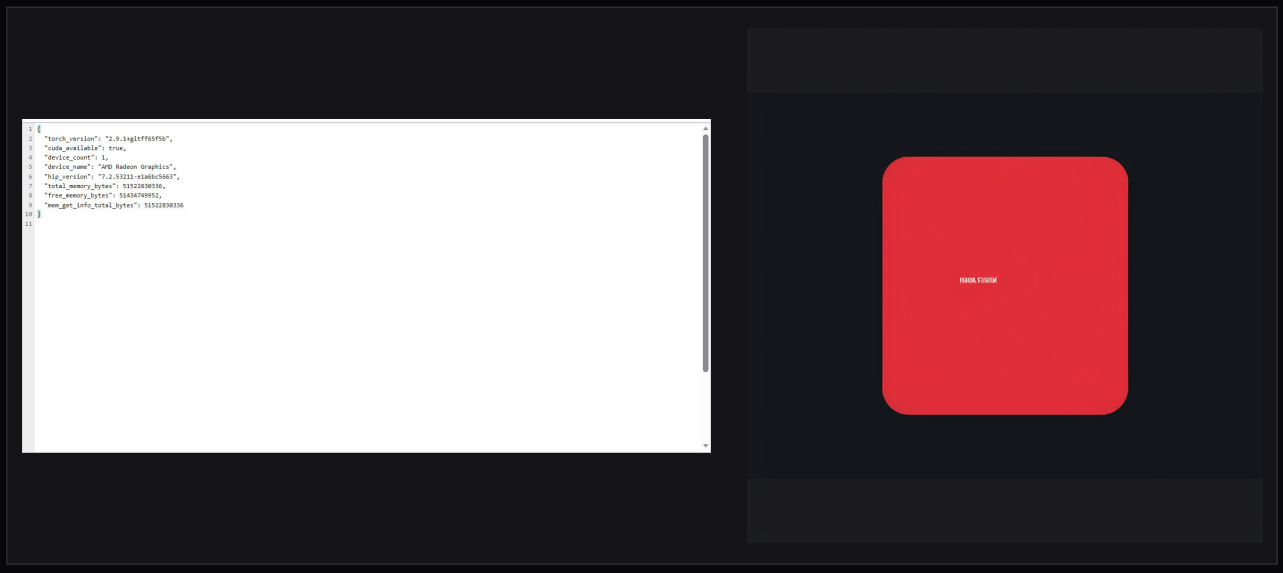
Architecture keeps every boundary visible



- Next.js frontend for responsive campaign configuration and review
- FastAPI for upload validation, JSON persistence, protected results, and review
- Experimental single-job worker with lazy load and pipeline reuse
- Pinned SDXL img2img in fp16 on AMD Radeon through PyTorch/HIP
- Bearer authentication, explicit CORS, loopback bind, and no CPU/mock fallback

RADEON IMPLEMENTATION

Real SDXL ran on AMD Radeon with ROCm



GPU / architecture	**AMD Radeon Graphics / gfx1100**
PyTorch / HIP	**2.9.1+gitff65f5b / 7.2.53211-e1a6bc5663**
Model	**SDXL base 1.0, exact revision pinned**
Pipeline	**StableDiffusionXLImg2ImgPipeline, fp16**
Runtime choices	**No CPU offload; no attention slicing**

The real image above was generated on AMD Radeon with ROCm during the verified cloud session.

VERIFIED RESULT

Measured performance, not projected performance

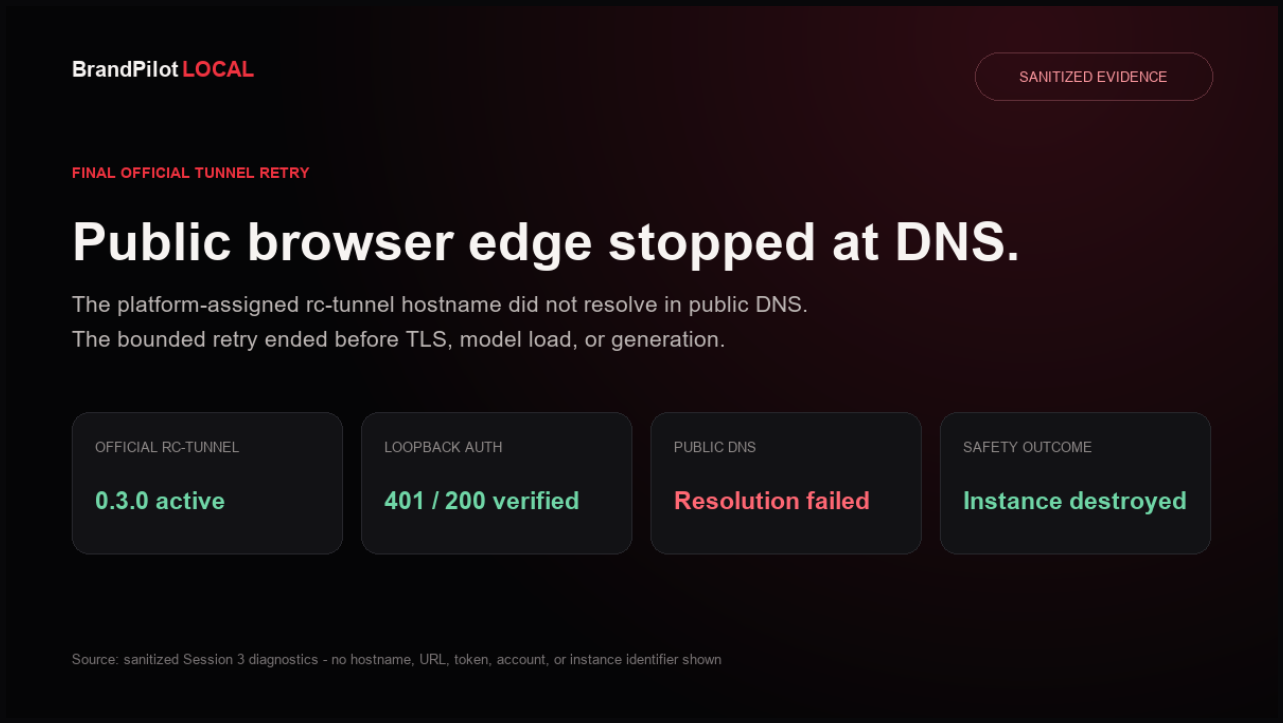
Resolution	**768 x 768**
Steps / strength / guidance	**15 / 0.35 / 6.5**
Seed	**20260725**
Cold model load	**132.295 seconds**
Generation	**222.111 seconds**
Total smoke test	**357.706 seconds**
Peak allocated / reserved	**13.41 GiB / 14.31 GiB**
Output SHA-256	**bce21d016a8179ac4ddabcbf5869a71eacf03565948d0d1f97d24e279b125452**

Application-integrated loopback proof

A later authenticated FastAPI-to-Radeon loopback job completed in 252.243 seconds from start to completion and persisted an Approved human-review state. Session 1 memory/hash values are not presented as measurements from that later job.

HONEST VALIDATION

The public edge is the one missing link

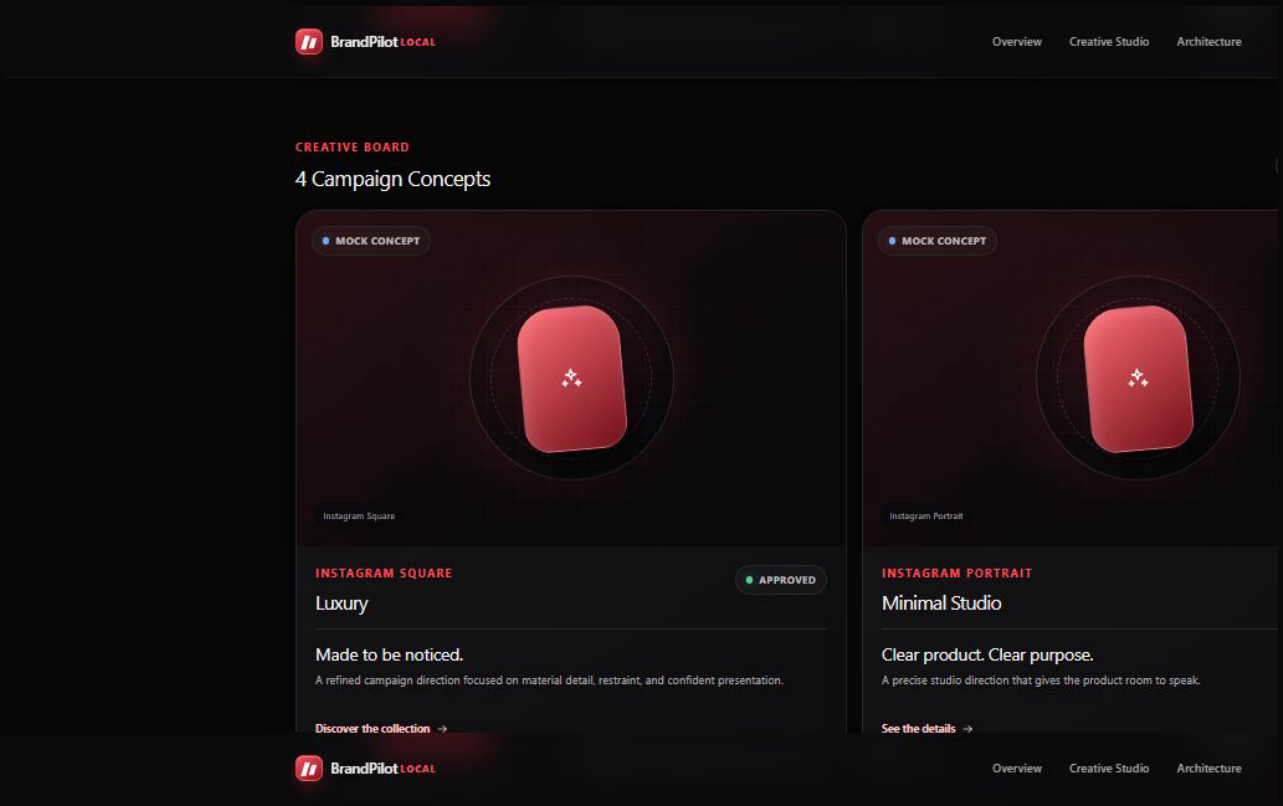


Standalone SDXL on Radeon/ROCm	**VERIFIED**
Authenticated FastAPI loopback generation	**VERIFIED**
Official rc-tunnel 0.3.0 install and activation	**VERIFIED**
Chrome TLS/auth/CORS/project/upload in Session 2b	**VERIFIED**
Protected real result rendered through public edge	**BLOCKED - DNS**

No unofficial tunnel, TLS bypass, host override, or fabricated success was used. The final retry stopped before model load and the instance was destroyed.

RESPONSIBLE AI

Human review is part of the result



- Every mock or experimental result keeps its disclosure.
- Approval, rejection, and requested changes are stored with the project.
- Rights-cleared inputs, generated-media disclosure, and product-claim review remain required.
- Local processing can reduce exposure; it does not replace copyright, trademark, or commercial-use review.

VALUE AND ROADMAP

Practical today, disciplined about tomorrow

BrandPilot's innovation is the accountable campaign system around generation: brand context, multiple directions, output framing, provenance, persistence, and human decision-making in one premium responsive product.

Current limitations

- No completed public browser-to-Radeon generation or permanent hosted backend
- No product identity, logo, or packaging-text preservation claim
- No IP-Adapter, ControlNet, LoRA, fine-tuning, video, or comparative benchmark
- No production readiness or commercial model/source clearance

Next measured phase

After submission, compare base SDXL against IP-Adapter plus masking/compositing using defined product-preservation quality criteria.

Team

Talal Khawaja	**Project lead, application development, Radeon integration, documentation, and demo preparation**
Aqeela Urooj	**Product testing, output review, usability feedback, and documentation review**

Repository: github.com/kmt9967/brandpilot-local

Evidence map: docs/judging-evidence-map.md

Claim boundary: docs/final-claim-audit.md